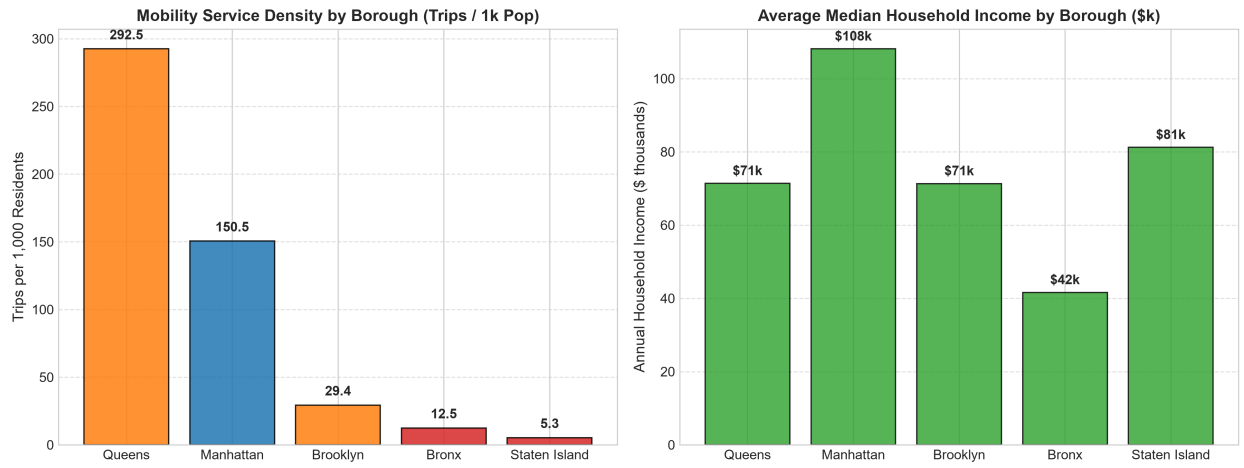
Figures

Geographic and Socioeconomic Bias in Spatial Mobility Data
Namith Kumar C (PES1UG24AM171) and Jayanth K (PES1UG24AM123)

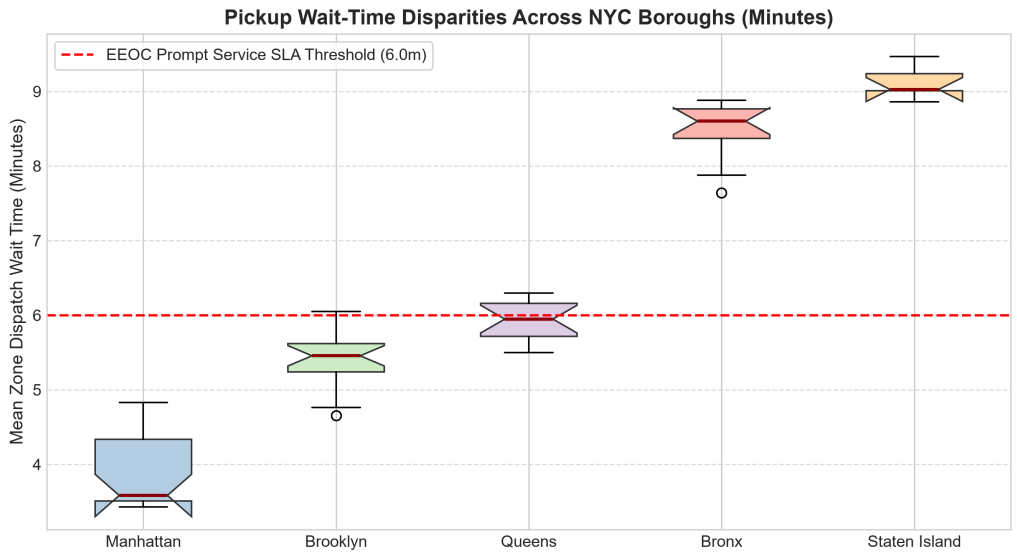
Figure 1	Service Density and Household Income by Borough
Figure 2	Pickup Wait-Time Disparities Across Boroughs
Figure 3	Lorenz Curve of Spatial Mobility Allocation
Figure 4	Median Income vs Service Wait Time
Figure 5	Policy Simulation Comparison
Figure 6	Pareto Frontier: Platform Efficiency vs Spatial Fairness

Figure 1. Service Density and Household Income by Borough



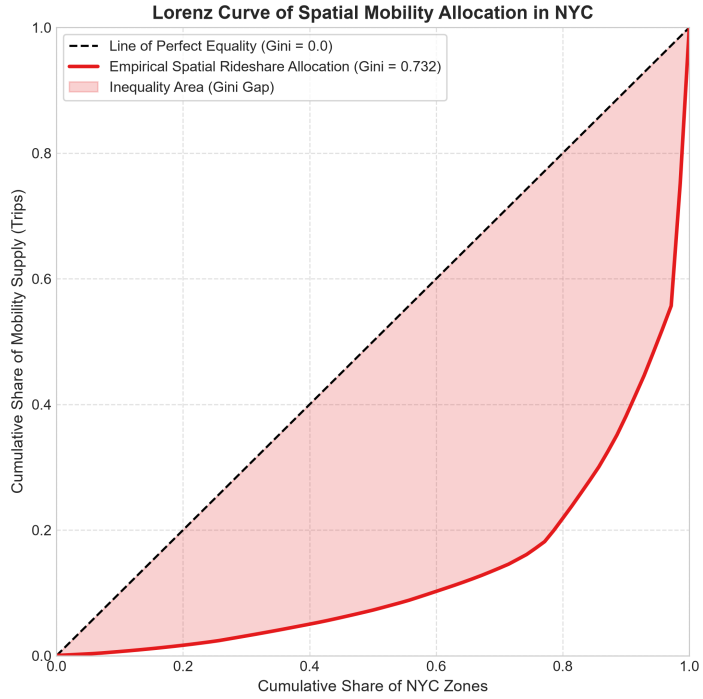
Left: trips per 1,000 residents (Queens 292.5, Manhattan 150.5, Brooklyn 29.4, Bronx 12.5, Staten Island 5.3). Right: average median household income in $ thousands (Manhattan 108, Staten Island 81, Queens 71, Brooklyn 71, Bronx 42). Queens is elevated by the JFK and LGA airport zones.

Figure 2. Pickup Wait-Time Disparities Across Boroughs



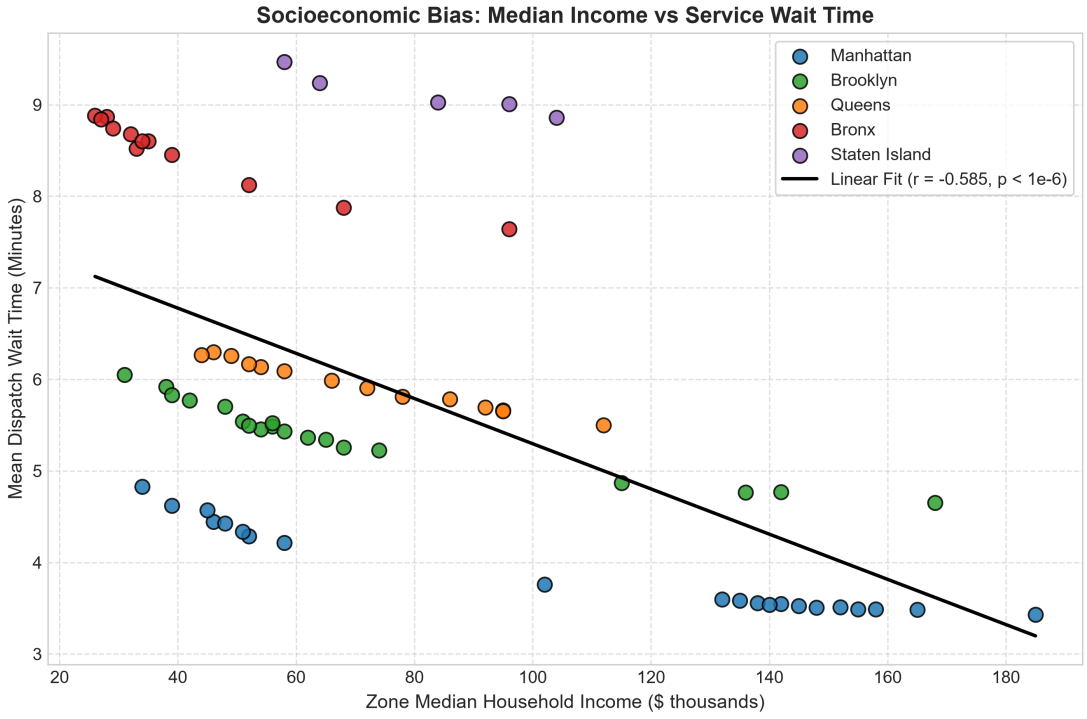
Notched box plots of mean zone dispatch wait time (minutes) per borough, with the 6.0 min service threshold shown as a dashed line. Manhattan and Brooklyn sit below the threshold, Queens straddles it, and the Bronx and Staten Island are well above it.

Figure 3. Lorenz Curve of Spatial Mobility Allocation



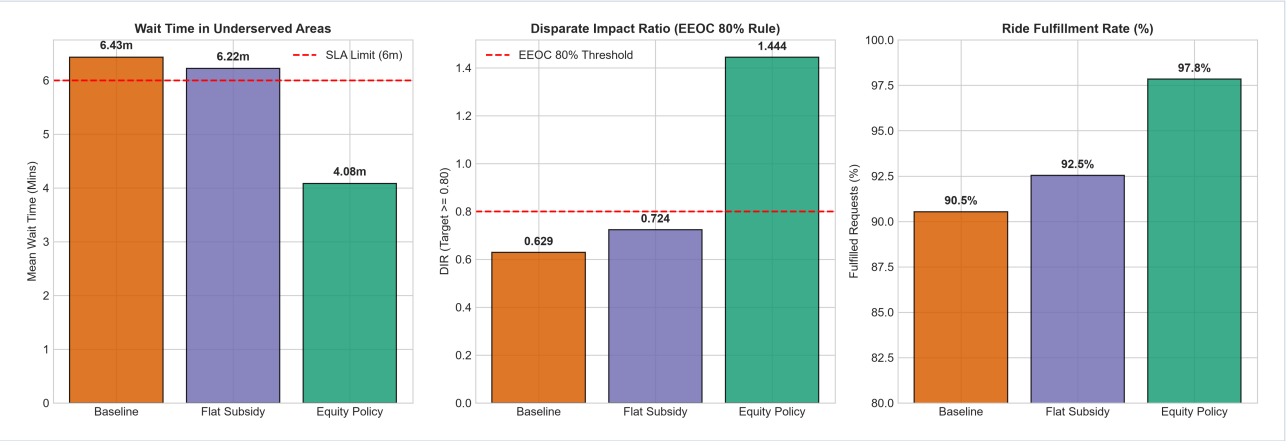
Cumulative share of trips against cumulative share of NYC zones. The shaded area between the line of perfect equality and the empirical curve corresponds to a Gini coefficient of 0.732.

Figure 4. Median Income vs Service Wait Time



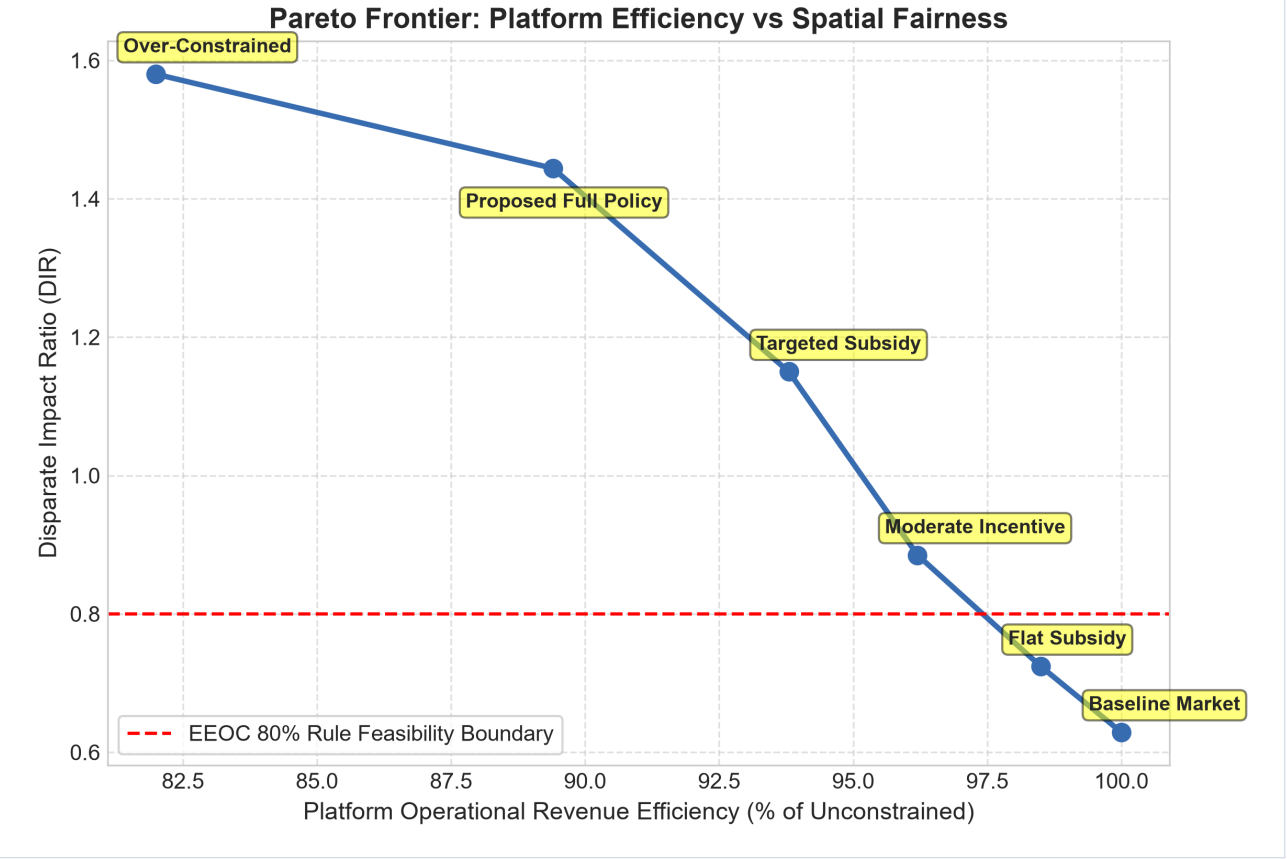
Each point is a zone, coloured by borough, with a linear fit (r = -0.585). Lower-income zones tend to wait longer, with clear borough-level clustering (Bronx and Staten Island high, Manhattan low).

Figure 5. Policy Simulation Comparison



Baseline, flat subsidy and equity policy compared on three measures. Wait time in underserved areas: 6.43 / 6.22 / 4.08 min. Disparate Impact Ratio: 0.629 / 0.724 / 1.444 (EEOC 80% threshold at 0.80). Ride fulfillment rate: 90.5% / 92.5% / 97.8%.

Figure 6. Pareto Frontier: Platform Efficiency vs Spatial Fairness



Disparate Impact Ratio against platform revenue efficiency (% of unconstrained) across policy settings, from the baseline market (100% efficiency, DIR about 0.63) to the over-constrained setting (about 82% efficiency, DIR about 1.58). The dashed line marks the 0.80 threshold.